# Sensor-controlled Conveyor Sorting

Task 3 | Automation system design | 3-page explanation

## Purpose and operating assumptions

This original design moves one discrete part at a time to a sorting station, stops the belt, pushes the part into a side chute, retracts the diverter and confirms exit before resuming. The part feeder must admit only one item into the controlled zone. The design uses a PLC-like scan model at 20 ms and ideal Boolean sensors. Settling time 0.20 s, cylinder travel timeout 1 s and exit timeout 2 s are adjustable design assumptions. They require measurement during physical commissioning.

| Input | Meaning |
|---|---|
| Start / Stop / Reset | Separate operator commands; rising edges used for Start and Reset |
| safe_ok / overload_ok | Healthy safety-chain feedback and motor protection |
| station / exit | Part at the diverter; fresh departure detection |
| extended / retracted | Cylinder end-position feedback |

| Output | Meaning |
|---|---|
| motor | Belt run command only in RUN |
| extend | Solenoid command only in EXTEND |
| alarm / fault reason | FAULT indication and human-readable reason |
| count | Increments once per confirmed departure |

## Design files

system_design.svg is an editable functional architecture diagram. flowchart.mmd is the editable Mermaid sequence. conveyor_controller.py is the executable behavioural design; test_conveyor.py exercises the sequence and fault paths. These satisfy the design-file requirement without claiming a vendor PLC project or production wiring design.

# Sequence, feedback and recovery

| State | Action and transition |
|---|---|
| IDLE | Outputs off. Fresh Start requires clear station/exit and valid retracted feedback. |
| RUN | Belt on. Station sensor -> SETTLE; belt turns off. |
| SETTLE | Wait 0.20 s before diverter activation. |
| EXTEND | Solenoid on. Extended feedback -> RETRACT; 1 s timeout -> FAULT. |
| RETRACT | Solenoid off; assumed spring return. Retracted feedback -> VERIFY; 1 s timeout -> FAULT. |
| VERIFY | Belt off. Fresh exit edge with station clear -> count +1, RUN; 2 s timeout -> FAULT. |
| FAULT | Outputs off. Clear cell, restore healthy feedback, retract cylinder and issue fresh Reset -> IDLE. A separate fresh Start is required. |

## Global interlocks

Loss of safe_ok or overload_ok immediately puts the model in FAULT and disables both commands. Contradictory extended/retracted feedback is a fault. Stop during an active sorting cycle, or with a part at the station, latches a fault so recovery cannot resume a partially completed sequence. Stop in a clear RUN state returns to IDLE. Holding Start through a fault and reset does not restart the belt.

## Safety architecture and practical selection

A real implementation needs independently selected safety equipment, guarding, drive stop functions and control of stored pneumatic energy. The ordinary PLC receives safety status for diagnostics; it does not replace the safety relay. De-energising a valve may retract the cylinder, which can itself create a hazard. Actual isolation and stopping behaviour must follow the application risk assessment. OSHA and manufacturer integration guidance support assessing the complete cell rather than relying on the robot or controller alone [2,9].

# Verification and implementation notes

## Executed software checks

Ten automated tests passed: normal sort and single count; safety loss in every active state; overload; extend timeout; retract timeout; exit timeout; held Start preventing restart; blocked Reset while the cell is occupied; contradictory cylinder feedback; and interrupted-cycle fault. Run python test_conveyor.py. The test transcript is included in validation.txt. These tests establish the coded behaviours under synthetic inputs; they do not prove a safety integrity level or physical reliability.

## Proposed physical commissioning

Confirm sensor polarity and supply compatibility; prove each input individually; test the diverter unloaded; measure extension and return times; tune settling and timeout thresholds; run representative parts; deliberately block a transfer under controlled conditions; verify diagnostics and reset. Check that departing parts actually cross the exit sensor and clear the station before counting. Establish a documented method to clear a jam with hazardous energy controlled. Record acceptance results separately from these simulation tests.

## Known boundaries

No feeder model, multi-part tracking, sensor debounce or network interface is implemented. A held station signal eventually prevents exit verification, but a missing arrival is indistinguishable from an empty line without additional sensing. The exit edge must occur in VERIFY; sensor positioning must ensure this in a build. The command sequence assumes a spring-return diverter, not a double-solenoid cylinder. No component part numbers, supplier quotations or measured throughput are invented.

## References


[2] OSHA. *Technical Manual, Section IV, Chapter 4: Industrial Robot Systems and Industrial Robot System Safety*. n.d.. Accessed 7 October 2026. https://www.osha.gov/otm/section-4-safety-hazards/chapter-4

[9] Universal Robots. *Risk Assessment, online manual SW10.15*. n.d.. Accessed 7 October 2026. https://www.universal-robots.com/manuals/EN/HTML/SW10_15/Content/prod-usr-man/complianceG7/H_g5_sections/safety_g5/risk_assessment.htm

[11] Python Software Foundation. *unittest: Unit testing framework*. n.d.. Accessed 7 October 2026. https://docs.python.org/3/library/unittest.html